

1) Employee Payroll System

START

1. Declare variables:
BasicSalary, HRA, Bonus, GrossSalary (as double / float)
 2. Read BasicSalary
 3. Read HRA
 4. Read Bonus
 5. Check for valid input:
IF BasicSalary < 0 OR HRA < 0 OR Bonus < 0 THEN
Display "Invalid Input: Values cannot be negative"
Else
Calculate GrossSalary:
$$\text{GrossSalary} \leftarrow \text{BasicSalary} + \text{HRA} + \text{Bonus}$$

Display GrossSalary.
END IF
- STOP

2) Employee Performance Evaluation

START

1. Declare Variables:

PerformanceScore (as integer / float)

2. Read PerformanceScore

3. Check edge cases and evaluation condition:

IF PerformanceScore < 0 OR PerformanceScore > 100 THEN

Display "Invalid Input: Performance Score must be between 0 and 100"

ELSE IF PerformanceScore $\geq$ 80 THEN

Display "Eligible For Incentive"

ELSE

Display "Not Eligible"

END IF

STOP

3) Shopping Bill calculator

START

1. Declare variables:

Quantity (as integer)

Price, DiscountPercentage, GSTPercentage, SubTotal,
DiscountAmount, TaxAmount, TotalAmount (as double / float)

2. Read Quantity

3. Read Price

4. Read DiscountPercentage

5. Read GSTPercentage

6. check for valid input:

IF Quantity ≤ 0 OR Price < 0 OR DiscountPercentage < 0 OR
GSTPercentage < 0 THEN

Display "Invalid Input"

ELSE

calculate SubTotal:

Subtotal $\leftarrow$ Quantity * Price

calculate DiscountAmount:

DiscountAmount $\leftarrow$ Subtotal * (DiscountPercentage / 100)

calculate TaxAmount (GST):

TaxAmount $\leftarrow$ (Subtotal - DiscountAmount) * (GSTPercentage / 100)

calculate TotalAmount using addition and subtraction operators:

TotalAmount $\leftarrow$ Subtotal - DiscountAmount + TaxAmount

Display TotalAmount

END IF

STOP

4) Student Scholarship System

START

1. Declare Variables:

Attendance, Marks (as float/double)

2. Read Attendance

3. Read Marks

4. Check input boundary conditions and evaluate using NESTED IF:

IF Attendance ≥ 0 AND Attendance ≤ 100 AND Marks ≥ 0
AND Marks ≤ 100 THEN

IF Attendance ≥ 75 THEN

IF Marks ≥ 85 THEN

Display "Eligible for Scholarship"

ELSE

Display "Not Eligible: Marks below 85%"

END IF

ELSE

Display "Not Eligible: Attendance below 75%"

END IF

ELSE

Display "Invalid Input: Attendance and Marks must be between 0 and 100"

END IF

STOP

5) Industrial Sensor Data Processing

START

1. Declare Variables:

RawTemperature (as float/double)

StoredTemperature (as integer)

2. Read RawTemperature

3. Convert floating-point temperature to integer

Using ~~Explicit~~ EXPLICIT TYPE CASTING:

StoredTemperature $\leftarrow$ static_cast (RawTemperature)

4. Display RawTemperature

5. Display StoredTemperature

STOP.